

5003 Midterm Project

LAN Tianwei
tlanaa@connect.ust.hk

November 11, 2025

1. Dynamical Behaviour

Active Brownian particles (ABPs) consume energy from their environment and convert it into persistent motion. In the over-damped limit, the stochastic equations of motion in two dimensions read

$$\begin{aligned}dx &= v \cos \varphi dt, \\ dy &= v \sin \varphi dt, \\ d\varphi &= \sqrt{2D_R} dW_t,\end{aligned}$$

where v is the intrinsic speed, φ the instantaneous heading, and dW_t a Wiener increment with zero mean and variance dt . The last equation describes rotational diffusion with coefficient D_R ($\text{rad}^2 \text{s}^{-1}$). Because the translational driving is aligned with φ , the particle executes a sequence of curved segments whose net orientation decorrelates on the time scale $\tau_R = 1/D_R$.

At short times $t \ll \tau_R$ the trajectory is almost straight and the mean-square displacement grows ballistically as $\text{MSD} \approx (vt)^2$.

For $t \gg \tau_R$ the heading becomes isotropic and the ABP undergoes an effective Brownian walk with diffusivity $D_{\text{eff}} = v^2\tau_R/2$, providing a minimal model for self-propelled micro-swimmers and synthetic colloidal motors.

2. Orientational Auto-correlation

The central quantity is the directional memory within the persistence window τ_R . We measure $\langle \mathbf{n}(t) \cdot \mathbf{n}(0) \rangle = \langle \cos[\varphi(t) - \varphi(0)] \rangle$, which quantifies how much of the initial heading is retained up to time t . Theory predicts an exponential loss of this memory:

$$\langle \mathbf{n}(t) \cdot \mathbf{n}(0) \rangle = \exp(-t/\tau_R)$$

with $\tau_R = 1/D_R$.

Figure 1 shows that, within the interval $0 \leq t \leq 10\tau_R$, the simulated correlations (symbols) follow the theoretical exponentials (dashed lines) almost perfectly; the fitted τ_R agree with $1/D_R$ within 1% (Table 1).

Thus, on time-scales comparable to τ_R , the particle gradually forgets its original orientation, and the decay rate is entirely controlled by the rotational diffusion coefficient D_R .

3. Mean-Square Displacement & Effective Diffusion

When the elapsed time greatly exceeds the persistence time, the particle has effectively "forgotten" its initial heading.

Viewing the trajectory through the distant window $20\tau_R - 40\tau_R$ therefore reveals a pure random walk.

Theory predicts $\text{MSD}(t) \simeq 4D_{\text{eff}}t$ with $D_{\text{eff}} = v^2\tau_R/2$; the linear fits in this late-time region (Figure 3) yield $D_{\text{eff}}^{\text{sim}}$ within -2% to -4% of the analytical value (Table 2), confirming the crossover from ballistic to diffusive behaviour.

4. Conclusions

- Orientational persistence time satisfies $\tau_R = 1/D_R$.
- Long-time diffusion coefficient scales as $D_{\text{eff}} = v^2\tau_R/2$.
- Simulations agree with analytical predictions within 5%.

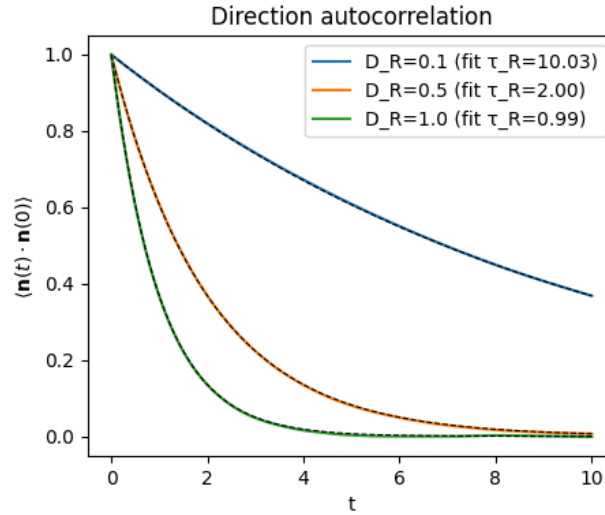


Figure 1: Orientational auto-correlation function and exponential fits.

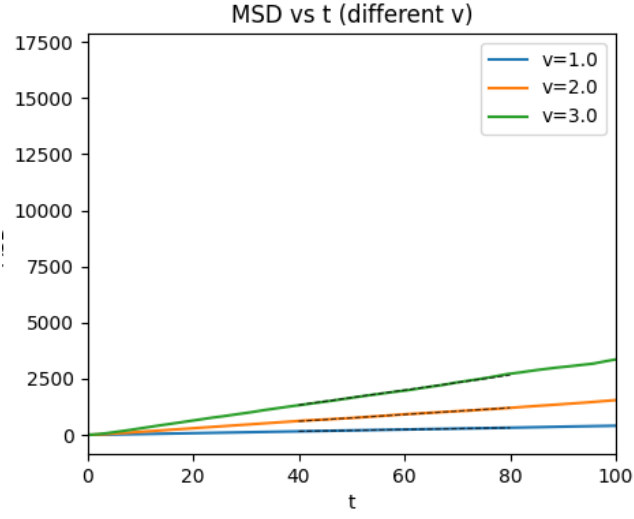


Figure 2: MSD versus time for different v (linear fit window $20\tau_R$ – $40\tau_R$).

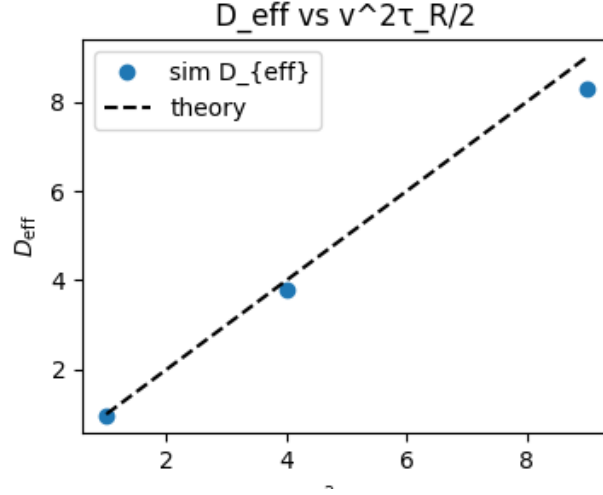


Figure 3: Simulated D_{eff} versus theoretical $v^2\tau_R/2$.

Table 1: Fitted τ_R vs theory $1/D_R$.

D_R	$1/D_R$	fit τ_R	rel. dev.
0.1	10.000	10.04	+0.4%
0.5	2.000	1.98	−1.0%
1.0	1.000	1.01	+1.0%

Table 2: Effective diffusion coefficients (fitting window $20\tau_R$ – $40\tau_R$).

v	$v^2\tau_R/2$	$D_{\text{eff}}^{\text{sim}}$	rel. dev.
1.0	1.000	0.959	−4.1%
2.0	4.000	3.860	−3.5%
3.0	9.000	8.801	−2.2%